

APOORV CHITNIS

Python & AI Engineer | Applied AI, Backend Systems & Cloud

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Python and AI engineer with enterprise experience in large-scale data validation, AI-assisted automation, and cloud workflows. Built production-oriented systems across Azure Databricks, LLM APIs, geospatial AI, optimization, and computer vision, with a current focus on agentic AI, RAG, evaluation, and deployment.

EXPERIENCE

MSCI | Analyst | Pune, India

Sep 2024 - Jun 2025

- Engineered a Python validation system for an Oracle-to-Azure Databricks migration spanning 50M+ records, using async I/O, multithreading, memory-efficient batching, YAML configuration, and Pytest to reduce projected validation runtime to under 24 hours.
- Built an AI-assisted UI/API automation proof of concept using LLM APIs to convert natural-language test cases into executable flows and adapt to dynamic DOM structures, cutting test-authoring effort by 90%+ in the prototype.
- Developed reusable validation for multi-stage SEBI data pipelines across Landing, Refined, and Distribution layers, covering Parquet/CSV transformations, schema evolution, rounding, and structural reshaping.
- Integrated validation suites with Azure DevOps CI/CD and secure connectors for Oracle, Cosmos DB, and Databricks; recognized internally for infrastructure-level impact and rapid upskilling.

SELECTED PROJECTS

HeatShift - Heat-Aware Operational Planning & Optimization

2026

- Built and deployed a full-stack planning system that combines location/date-aware environmental conditions, fictional large-scale site operations, heat-risk metrics, and deterministic constrained scheduling to restructure a workday while preserving hard operational constraints.
- Implemented a lexicographic optimizer over 30-minute candidate starts and eligible crews to minimize high-risk worker-minutes, total crew exposure load, peak individual exposure, schedule shifts, and crew reassignments.
- Benchmarked the screening score against 566 controlled HEAT-SHIELD sessions, observing a 0.7718 Spearman correlation with measured one-hour physical-work-capacity loss; documented limitations and safety boundaries. Stack: FastAPI, Next.js/React, Supabase, Leaflet, Vercel.

NASA Space Apps Challenge - GeoAI Reimagined

2023

- Built a real-time NASA HLS satellite-data fetcher and explored disaster surveillance using the Prithvi-100 geospatial foundation model with limited imagery.

AI-Based Crop Management System Using Drones - Patented

2023

- Worked on drone assembly/configuration and YOLOv5-based vision for crop monitoring and selective orchard fogging; project won HackMIT-WPU's Computer Science Innovation prize and resulted in Design Patent 404133-001.

SKILLS

Languages: Python, SQL, C++, C

AI / ML: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, LangChain, MLflow, LLM APIs, RAG, tool calling, agentic workflows, MCP

Backend / Data: FastAPI, Flask, Django, Pydantic, Pandas, NumPy, Pytest, PostgreSQL/Supabase, MySQL, OracleDB

Cloud / DevOps: Azure, AWS, Azure Databricks, Docker, Azure DevOps, Git/GitHub, Vercel, CI/CD

EDUCATION & SELECTED ACHIEVEMENTS

MIT World Peace University, Pune | B.Tech. Computer Science Engineering | CGPA: 9.1/10

2020 - 2024

- Publication: "Using Drone Technology for Fruit Orchard Management and Waste Reduction," ICCUBEA.
- Design Patent 404133-001 - AI-based Crop Management System using Drones; HackMIT-WPU Innovation Project winner; Flow Blockchain Hackathon Best Presentation.